SoK: Reusing Defense Evidence in LLM and Agent Side Channels

***Abstract*—Choosing a side-channel defense requires knowing which reported outcomes apply to the intended deployment. This SoK organizes that evidence for LLM serving and agent systems. We map 82 study families and examine 39 result records, including 33 mitigation records from 20 families. A taxonomy of changes to the observed signal, executed intervention and evaluation setting guides three practical tasks: compare like outcomes, identify protection, utility and cost measured at the same configuration, and specify what a new deployment still needs to measure. A running cache-reuse example shows the distinction: Early Bird's prefix-recovery result and semantic-cache overhead concern different mechanisms, whereas CachePrune links two recovery targets to one quality-and-cost package. We publish the record identities and query memberships needed to reconstruct these decisions. Pair counts are an inventory conditioned on record granularity, not an estimate of deficient research. Worked applications retain useful source results while separating incompatible cost or utility claims. The resulting evidence atlas and reporting template support source-scoped candidate selection and explicit measurement plans.**

1. Introduction

Encryption conceals message content, yet LLM serving and agents expose packet sizes, cache timing, destinations and tool sequences [1], [2], [3], [4], [5], [6]. Padding, cache isolation and decoy actions address different parts of this problem. A practitioner must decide which reported protection, quality and overhead measurements apply to the system they intend to deploy.

One example, three decisions. Consider a serving operator choosing a cache-reuse defense. Early Bird evaluates a prefix-recovery oracle and separately measures semantic-cache deidentification overhead [3, V-A–B]. CachePrune evaluates direct and contextual recovery and links its default configuration to generation quality and serving cost [7, 5.1–5.6]. First, prefix and direct-recovery results can inform a shortlist for a shared token-recovery threat. Second, Early Bird's semantic-cache overhead cannot price the prefix policy, while CachePrune's shared package can inform feasibility at its tested configuration. Third, a changed workload or serving system needs the corresponding workload and cost measurements. The original papers need not have asked the operator's deployment question: the task is to reuse their results without changing what those results measure.

This SoK develops that task across LLM and agent side channels. We distinguish changes to the *observed signal*, the *executed intervention*, and the *evaluation setting*. These categories connect low-level isolation and traffic shaping with a newer complication: a cover action can alter an agent's environment, and a component's output quality can differ from a planner's task success. The 82-family map supplies the landscape. The defense synthesis is deliberately bounded to 33 mitigation records from 20 families, with six contextual records used to explain observation and execution boundaries.

Contributions.

- **Comparison rules for scoped results.** We organize changes of signal, intervention and setting into explicit comparison questions (§§3–5). The complete pair inventory includes source locators, reasons and sensitivity to record granularity. Its counts document the indexing choices; the useful output is the permission to retain a particular comparison.
- **An atlas of outcomes measured together.** The record audit identifies which protection, utility and cost measurements share an evaluated configuration. Main-text identities and same-PDF query memberships let a reader reconstruct each supported combination and distinguish it from a union of family-level cells (§5).
- **Deployment measurement plans.** The holdout synthesis separates attack generalization from tested defense transfer. A reusable reporting card and filled examples translate a proposed deployment change into retained evidence and a specific next measurement (§§6–8.1).

The synthesis builds on website-fingerprinting and adaptive-attack evaluation practice [8], [9], [10], [11]. Its added organizing question is which *executed object and outcome* each measurement describes when serving primitives, tool state and planner utility coexist. External applications and internal examples show how to fill the evidence atlas and reporting card; they do not supply a comparative accuracy estimate for the procedure.

Scope of the synthesis. The mapped collection supplies context; evaluated mitigation records supply defense evidence. A missing report limits the evidence available for a specified reuse question, without establishing that the original study omitted an experiment or failed at its own objective.

TABLE 1. RESEARCH QUESTIONS, SYNTHESIS OUTPUTS AND THEIR SCOPE.

Question	Synthesis output	Concrete use / evidential basis
RQ1: landscape	Map hidden targets and signal-generating versus receiving layers.	82 study families supply context; 20 families supply the evaluated mitigation records.
RQ2: comparability	Retain like targets while separating changed mechanisms, execution modes and cost objects.	Early Bird prefix recovery and CachePrune direct recovery form candidates; semantic-cache overhead does not price prefix isolation. Pair counts depend on granularity.
RQ3: joint evidence	Identify outcomes measured at the same configuration, with all query members listed.	CachePrune gives one source-scoped quality/cost package for two targets; SparLeak supplies intensity-specific inference/quality pairs.
RQ4: transfer	Distinguish attack generalization from a tested defense or transported calibration.	The three model and two topology holdouts test attack generalization; a proposed defense extension requires a linked defended condition.

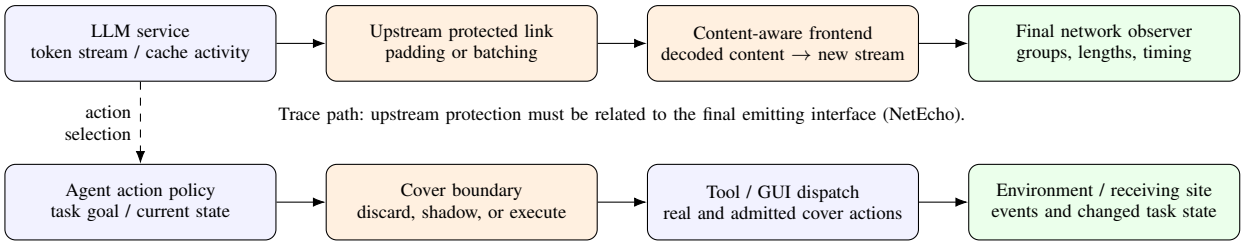


Figure 1. Receiving views and execution boundaries. A content-aware frontend can emit a new trace; admitted cover actions can change state. Evidence must describe the object a deployment will expose or execute.

2. Scope and Systematization Method

2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions. **RQ1 (landscape):** which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems, and where is each signal generated versus observed? **RQ2 (comparability):** when do two reported defense results measure the same target, receiver, intervention and cost object? **RQ3 (joint evidence):** which results bind protection, utility and cost at one evaluated operating point? **RQ4 (transfer):** which evidence distinguishes attack generalization, defended-condition evaluation and defense transfer across tasks, models, topologies and hardware? RQ1 supplies the field context; RQ2–RQ4 correspond to the three contributions.

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [12], [13], [14], [15], [16]. Table 2 states the observable eligibility rules. For C3, the extraction must name an execution mechanism and an evaluated observable separate from ordinary linguistic meaning. A filter qualifies when its accept/reject behavior or differential response is the measured oracle for a hidden property; merely analyzing a reply's wording or style does not. If a mixed paper does not isolate that mechanism, its membership stays adjacent. This operational distinction concerns the measured signal, not whether the output is text.

TABLE 2. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Core only for an isolated accept/reject or differential-response oracle tied to filter execution; semantic content alone remains adjacent.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized

identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest: **Type I** concerns confidentiality/privacy; **Type II** concerns model, harness or identity attribution; **Type III** concerns evasion of a platform's human–agent detector; **Type IV** concerns authorized operator diagnostics. Types can overlap. WRA protects research intent (I); UI Traces attributes agents (II); AHB evades a defensive detector (III); ClawGuard diagnoses loops (IV) [6], [5], [17], [18]. These roles are mapped per core family, alongside generating and receiving layers, in Table 7.

2.2. Corpus construction and reconciliation

Forty seeds, venue-targeted proceedings screening and one-hop citation snowballing produced an initial map of 60 families (43 core, 17 adjacent). A supplementary database and bulk keyword search returned 615 occurrences and 565 unique records; auditing its 204 exclusions rerouted four records, and the search added 15 core and seven adjacent families. The final collection has 58 core and 24 adjacent families. The completed search frames, closed on October 21, 2026, combine Semantic Scholar and native arXiv queries with venue-targeted screening of S&P, USENIX Security, NDSS, CCS and PMLR. C1–C5 screening and version deduplication operate on records before family reconciliation; the supplement retains queries, dispositions and excluded incomplete search frames.

2.3. Source-bound result records

Each record links its target, observation boundary and operating point to a source version and section/table/figure locator. The authors report reviewing the classification of all 82 families using titles, abstracts and experimental sections, and checking all 39 result records and 180 family-audit cells against source locators. The subsequent 351-cell record audit and 133 within-question pair decisions retain author-assigned reasons; executable checks establish their accounting consistency. Candidate fields produced with AI assistance were subject to this author review. The historical material does not provide a complete item-linked draft-to-final error log, so it supplies no calibrated AI extraction-error rate. Thirteen later changes corrected the family aggregation/display while leaving record judgments fixed; they are not 13 measured extraction errors.

Appendices B and F contain the record index, comparison summary and query memberships in this PDF. The ledger also separates cited publication status, inspected version, measurement mode and remaining binding uncertainty. Published status increases traceability of venue review but does not establish the correctness of a result; a preprint's observations remain provisional at their inspected scope.

SoK Takeaway 1 (Map evidence to the receiving deployment). A family-level label does not identify which signal, observer permission or protected objective a result supports. The complete core map separates signal origin from receiving access; the result ledger then binds each comparison to a source version and operating point.

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

3. Result Records and Comparability Rules

The atomic unit of our systematization is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration (D_0: clean-condition fit; D_2: protected-condition fit), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A record is a comparison unit with a source-indexed parameter grid, not an independent trial. We split when a change of primitive, execution mode, target/receiver contract or separately measured cost object would alter what a result can support. We retain models, datasets, parameter values and repetitions as coordinates within that unit; numerical joins must preserve those coordinates. Source versions are reconciled within a family before extraction.

Indexing schema and comparison contracts. The index names the hidden property, measured channel, observer permissions, intervention and evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (U), unavailable (A) or unverified (?). Table 5 illustrates a complete record-to-judgment lineage.

Record boundaries and coverage. The extraction frame comprises the evaluated-mitigation sections and their linked methods, figures and cost/utility analyses for the 20 mitigation families, plus six explicitly named contextual controls. A defense discussion without a scoped evaluated

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	Dangol workflow / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [19].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about $13\times$ bytes and $11\times$ latency in that condition.
Unresolved fields	Objective terminal task utility U; precise paired live-refit comparison ?.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

result stays in the map. The 39-record ledger inventories these comparison units, rather than every plotted number in the 82-family corpus. S34 publishes all record scopes and split decisions. Three examples fix the granularity:

- **Split:** Early Bird's K-prefix recovery oracle and semantic-cache processing benchmark change both primitive and measured object; their numbers cannot form one protection–cost point.
- **Retain coordinates:** SparLeak's model-by-replacement-intensity ASR/ROUGE grid remains one mitigation record. Each intensity's inference/quality pair must be retained together; selecting the lowest ASR and highest ROUGE from different rows is disallowed.
- **Split outcome, share package:** CachePrune's direct recovery and contextual guessing require two target contracts. They reference one shared default quality/cost package, counted once when joining configurations.

TABLE 6. TAXONOMY OF CHANGES RELEVANT TO EVIDENCE REUSE.

Changed object	Typical transition	Evidence to carry or renew
Observed signal	Receiver, released representation, hidden target	Keep the target and permissions; measure the actual receiving trace.
Executed intervention	Prefix/semantic cache; transformed trace/live scheduler; read/write cover	Keep mechanism and state boundary; bind function and cost to the executed configuration.
Evaluation setting	Attacker fit, workload, model, hardware or network	Keep training/test provenance; establish the proposed holdout and relevant baseline.

Question cohorts precede compatibility judgments. Records enter one declared comparison question by the intervention and receiving object, not by score or verdict: delivered traffic, service-cache reuse, local memory/resource behavior, behavioral humanization, generator fingerprinting or fixed instructions. Three contextual singletons concern access gates, contention and filter-created views. S34 lists operational membership rules and all 39 assignments. When a mechanism spans several questions, retain the primary extracted question and document the alternative; a new cross-cohort question requires a new inventory. A shared cohort admits a pair for inspection; it does not make different targets or costs compatible. These retrospective grouping rules define the inventory. Splitting records on changed objects and then observing different objects is partly a consequence of that definition. We therefore use pair labels to delimit comparison permissions, not to estimate defects or the prevalence of incomparable research. Section 5 varies the unit explicitly.

A taxonomy of reuse changes. The categories below describe what changes between a source result and a deployment question; several can apply together. They organize the synthesis, while the nine audit items are the evidence needed to answer the resulting questions.

From contracts to an audit. Section 5 operationalizes these contracts as nine reuse items with a shared evidence-status alphabet (Table 11) and applies them to every mitigation family (Table 12).

4. Corpus Map Across 82 Study Families

4.1. Generating layers, receiving projections and threat polarities

Table 7 maps all 58 core families by signal-generating layers, receiving projections and threat polarity. Multi-layer families retain every reported layer in their generating and observing sets.

Cross-layer synthesis. A remote network observer, a same-site script and a privileged diagnostic hook expose different views even when they share an attribution target. WRA's read-only privacy decoys, AHB's stateful evasion

TABLE 7. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; LAYER AND TYPE LABELS MAY BE MULTIPLE AND DO NOT IMPLY ONE-TO-ONE CAUSAL PATHS. THE BLOCKS GROUP SOURCE LAYERS, NOT EVIDENTIAL QUALITY.

Core family	$G \to O$	Type
hardware generation, P1 (8)		
Edge EM leakage [20]	$1 \to 1$	I
Energon [21]	$1 \to 1$	II
MoEcho [22]	$1 \to 1$	I
I Know What You Said [23]	$1 \to 1$	I
Behind Bars [24]	$1 \to 1$	I
PAGE404 [25]	$1 \to 1$	I
Spill the Beans [26]	$1 \to 1$	I
ArchSnoop [27]	$1 \to 1$	II
hardware and service, P1/P2 (8)		
SparLeak [28]	$1,2 \to 1$	I
Shape and Substance [29]	$1,2 \to 1$	I
TDXRay [30]	$1,2 \to 1$	I
SparSEEty [31]	$1,2 \to 1$	I
OTRO [32]	$1,2 \to 1$	I
Detokenization Leaks [33]	$1,2 \to 1$	I
LLM-WFIN [34]	$1,2 \to 1,3$	I
LLM-serving RCA [35]	$1,2 \to 2$	IV
service generation, P2 (13 of 22)		
SplitDecoy [36]	$2 \to 2$	I
KVGov [37]	$2 \to 2,4$	I
LLMs Have Rhythm [38]	$2 \to 3$	II
Remote timing [39]	$2 \to 3$	I
NetEcho [40]	$2 \to 3$	I
Prompt Echo [41]	$2 \to 3$	I
Speculative decoding [42]	$2 \to 3$	I
Remote keylogging [1]	$2 \to 3$	I
Whisper Leak [43]	$2 \to 3$	I
Cache contention [44]	$2 \to 4$	I
Routing metadata [45]	$2 \to 4$	I
SafeKV [46]	$2 \to 4$	I
ML privacy channels [47]	$2 \to 4$	I

Core family	$G \to O$	Type
service generation, P2 (9)		
Early Bird [3]	$2 \to 4$	I
Prompt-cache audit [48]	$2 \to 4$	I
InputSnatch [49]	$2 \to 4$	I
Leaky Language Models [50]	$2 \to 4$	I
PromptPeek [2]	$2 \to 4$	I
SpliceLeak [51]	$2 \to 4$	I
Time Will Tell [52]	$2 \to 4$	I
CachePrune [7]	$2 \to 4$	I
MoE prompt stealing [53]	$2 \to 4$	I
cross-layer sources (5)		
ClawGuard [18]	$1,5 \to 1$	IV
MIRAGE [54]	$2,5 \to 2$	IV
Cloud traffic benchmark [55]	$2,5,8 \to 3$	I,II
Web-agent unmasking [56]	$3,7 \to 3,7$	II,III
LumiMAS [57]	$4,5 \to 4,6$	IV
orchestration/workflow sources (15)		
AgentPrint [4]	$5 \to 3$	I,II
Cycle detection [58]	$5 \to 4,6$	IV
SIGLEAK [59]	$5,6 \to 4,6$	I
ProcGrep [60]	$5,6 \to 4,6$	II
AgentProv [61]	$5,6 \to 6$	II
Lidar [62]	$5,6 \to 6$	II
Dangol workflow [19]	$5,6,8 \to 3,6,8$	I
Campaign attribution [63]	$5,6,8 \to 6,8$	II
WRA [6]	$5,7 \to 3$	I
PersonaFingerprint [64]	$5,7 \to 3$	I
Whose Agent? [65]	$5,7 \to 7$	II
AgentTell [66]	$5,7 \to 7$	I
UI Traces [5]	$5,7 \to 7$	II
FP-Agent [67]	$5,7 \to 7$	II
AHB [17]	$5,7 \to 7$	III

G denotes signal-generation layers and O the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and normalization of prose-only layer descriptions are in the artifact.

and SplitDecoy's exposed return path illustrate why receiving permissions and protected objectives must accompany the generating layer [6], [17], [36].

Layer incidence highlights where the systematization's comparison problem arises. In the 58-family map, signal-generation labels cluster at P2 serving/runtime (32 families) and P5 orchestration (19); receiving views cluster at P4 application/API (18), P1 hardware (16), and P3 transport (14). Generation at P3/P4 is rare (one family each), despite frequent observation at those layers. This source/receiver asymmetry makes the transfer question concrete: protection at a frequent source layer cannot be presumed to cover a different receiving projection. The map records layer sets, so it does not assign a causal edge to each source/receiver pair.

Purpose labels further show why a single "agent side channel" category is insufficient: Type I appears 41 times, Type II 13, Type III 2, and Type IV 5. The 32 map-only families extend coverage beyond mitigation comparisons: 15 list serving origins, 11 hardware and 11 orchestration; 19 concern confidentiality, nine attribution, one evasion and all five authorized-diagnostic families. Mapping these families locates privacy targets, operator capabilities and candidate validation environments even when no scoped mitigation record enters the ledger. Counts are overlapping family-label incidences; the audit then applies its reuse rules to the 39 scoped result records.

4.2. Relationship to prior systematizations

Prior surveys map compound-system threats, TEE boundaries and privacy contracts [68], [69], [70]. WF already supplies feature leakage analysis, cost trade-offs and environment-shift tests [71], [8], [72], [73]. Mathews et al. re-evaluate nine defenses with stronger attacks and an on-network DynaFlow implementation [9]. Adaptive attacks and online-cost evaluation are established foundations.

Adaptive evaluation and benchmarking. Tramèr et al. show why defense-specific adaptive attacks require more than applying an unchanged attack; Carlini et al. set out robustness-evaluation practices; RobustBench standardizes the evaluated threat and test interface [10], [11], [74]. These works already motivate protected-data evaluation and reproducible benchmarks. Our R4/R5 items instantiate that discipline. The LLM/agent-specific question is whether a refit, cost or functional result belongs to the same prefix policy, content-aware emitter or state-changing action under discussion.

What this synthesis adds. Our unit is a *scoped result*, so execution mode, protected state and cost remain attached when one paper reports several mechanisms. The 39-record atlas documents these distinct measurement objects; the 82-family holdout matrix separates evidence for each transfer axis. This connects evidence that earlier taxonomies organize by attack or mechanism: one family can contain several execution modes, a receiving projection can cross system layers, and utility/cost can belong to a different operating point from inference. The synthesis preserves these distinctions when translating a reported result into a deployment decision.

Formal guarantees and runtime measurements. The mitigation inventory covers evaluated runtime results, not the complete literature on privacy proofs. Differential privacy constrains releases for a specified adjacency relation and privacy budget [75]; information-flow measures distinguish mutual information from single-guess vulnerability [76]. Constant-time verification and oblivious memory protect stated traces under their own leakage models [77], [78]. These guarantees can supply stronger source-scoped evidence than a low empirical attack score. Their scope still matters: a guarantee for model weights need not include a serving trace, and hiding addresses need not hide access counts. TDXRay and OTRO connect these foundations to the indexed LLM components. For reuse, retain the stated adjacency, leakage model and execution assumptions. Attack-refitting items are inapplicable to a purely formal proof; absence of a classifier experiment does not count against that guarantee.

5. Record Comparability and Joint Defense Evidence

5.1. Unified 58-core disposition and 39-record ledger

The corpus partitions as $82 = 58$ core $+24$ adjacent families. Within the core, $58 = 20$ mitigation $+6$ contextual $+32$ map-only families. The ledger holds 39 records: 33 from the 20 mitigation families and one from each contextual family—SIGLEAK, AgentTell, FP-Agent, Behind Bars, LLMs Have Rhythm and ML privacy channels [59], [66], [67], [24], [38], [47]—which supply fixed-guardrail, access-control, interference, VPN or filter-origin contrasts rather than an audited mitigation claim. The 32 map-only families comprise 22 without a scoped mitigation evaluation, five diagnostics, four unbound quantitative assertions and SplitDecoy's boundary case. Table 8 groups mechanisms; its eight rows are mechanism groups, not sampled studies.

Whole-ledger synthesis. Each record has one primary measurement mode: 20 executed collections/prototypes, six trace transformations, five simulations/derived estimates, six component/calibration measurements and two unresolved placements (Table 21). Eleven of the 20 mitigation families contribute more than one record, and nine of them span more than one mode; this is why a mechanism name cannot carry utility or cost between records. Executed collection establishes its stated execution scope, not automatically full-workflow utility.

5.2. Comparison permissions and their sensitivity to granularity

The declared question groups generate 133 comparisons from 741 possible record pairs; 608 have no proposed comparison. Table 9 reports every within-question decision. A descriptive match permits a source-qualified shortlist or discussion of the same target, while retaining different costs and populations; it does not permit effect pooling. For example, Early Bird's prefix oracle and CachePrune's direct recovery can both inform a token-recovery shortlist. They do not share a measured serving-cost population.

What the mismatch count means. The 105 different-object pairs and eight of 15 same-family pairs are conditional outputs of the split and grouping rules. Some differences are the reason the records were split. We accordingly make no claim that these fractions reveal an empirical failure rate. Distinct targets, hardware, operating points and research objectives are normal design choices. The avoidable mistake belongs to a *reuse operation*: attaching Early Bird's semantic-cache cost to its prefix oracle, or attaching an offline timing transform's cost to a live scheduler. The inventory does not establish that the source authors made that mistake, and does not require every defense study to be mutually comparable.

What changes when records are merged. Table 10 reprojects the fixed decisions using three coarser units. A projected pair is unanimous only if all constituent judgments agree; otherwise it remains mixed. Family-only grouping hides all 15 within-family pairs and yields 44 cross-family pairs, eight mixed. Merging the two CachePrune targets at their shared package yields 38 units and 124 pairs. These are indexing sensitivities, not newly adjudicated studies; a coarser unit can conceal a difference without resolving it. The original 17/133 is therefore not assigned a "high/low" interpretation against a random baseline. Randomly permuting question membership would mostly create unjudged cross-question pairs and cannot provide a null distribution for validity.

Three useful outputs. A descriptive match retains scoped alternatives. A documented object change requires a measurement at the proposed new object. An unresolved link requires source clarification. This division tells a practitioner

TABLE 8. EIGHT INTERVENTION FAMILIES ORGANIZED BY THREAT POLARITY (TYPE I: CONFIDENTIALITY PROTECTION; TYPE II: PROVENANCE/IDENTITY ATTRIBUTION; TYPE III: BOT-DETECTION EVASION), RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. BECAUSE OBJECTIVES AND RISK MAGNITUDES DIFFER, THESE ROWS ARE NOT A COMMON EFFICACY RANKING.

Intervention & polarity	Evaluated view, target and source	Established result and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [40], [43].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains U.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [19].	The co-located loopback setting yields 0.30 recovery at about 13× bytes and 11× latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are U.
Action humanization (Type III: bot-evasion)	Mobile GUI touch traces; evading human–agent bot detection. AHB [17].	Unlike WRA's read-only privacy decoys, AHB's `Fake` mode dispatches live mobile GUI touches to evade platform bot detectors, dropping Trip Planning success from 0.75 to 0.15. Non-Fake `History+Long` (0.7143 success) avoids GUI mutation.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [46], [37], [7].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median 1.3 μs prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [51].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and $<$ 5ms/$<$ 1% prefill time; this does not price combined deployment.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [5].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [30], [23], [32].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

TABLE 9. ALL 133 WITHIN-QUESTION DECISIONS; 608 OTHER PAIRS HAVE NO DECLARED COMPARISON. COUNTS DESCRIBE THIS INVENTORY AND ITS GROUPING RULES.

Question	Match	Different object	Unresolved
behavioral humanization	1	2	0
cache service	2	37	6
fixed instruction	0	0	1
generator fingerprint	3	0	0
local memory	0	14	1
traffic envelope	11	52	3
Total	17	105	11

TABLE 10. SENSITIVITY TO THE COMPARISON UNIT. C/M/U: UNANIMOUS DESCRIPTIVE MATCH/OBJECT MISMATCH/UNRESOLVED; MIX: CONSTITUENT JUDGMENTS DIFFER. HIDDEN: ORIGINAL WITHIN-QUESTION PAIRS ABSORBED INSIDE A UNIT. NO NEW PAIR IS ADJUDICATED.

Unit	Units	C	M	U	Mix	Hidden
Scoped result records	39	17	105	11	0	0
Shared configuration package	38	16	98	9	1	1
Family and measurement mode	35	10	83	9	1	4
Family only	26	2	31	3	8	15

what to do next without treating a normal difference of research scope as a defect. Appendix F and the machine-readable pair inventory expose membership and reasons; family-deletion counts in S33 are dependency diagnostics only.

5.3. Nine-item reuse audit across 20 mitigation families

R1–R9 are author-defined reuse conditions, not a quality score. Table 11 links requested conclusions to necessary evidence. Missing evidence can block an analyst's deployment extension while leaving the source's own claim intact. Component utility suffices for component claims; R5 applies to adaptation comparisons. The current family matrix is generated from the 351 record–item judgments. Each item retains B if any scoped record supplies B, otherwise P, ?, U, A, then inapplicable, in that order. This availability index retains the supporting record IDs; it does not join different configurations. U means no binding located in the inspected text, including linked methods and results, rather than evidence that an experiment was never performed. The source reread and all prior codes remain archived; unresolved record-specific bindings stay visible in the projection.

TABLE 11. CLAIM-CONDITIONED REUSE RULES. EACH ROW NAMES THE REQUESTED INFERENCE AND THE EVIDENCE NEEDED TO SUPPORT IT.

Item	Reuse query → required binding (**B**)
R1 Target	Same protected interest → explicit hidden property, inference goal and authorization relation.
R2 Receiver	Same observer protection → permissions, released view and auxiliary inputs.
R3 Intervention	Same executed mechanism → placement/mode: stored data, wire emission or actions.
R4 Protected fit	Resistance to protected-data fitting → identified protected training/calibration and testing.
R5 Matched adaptation	Effect of adaptation → clean/protected fits tested on the same defended population.
R6 Workload	Changed workload/system → tasks, configuration and the relevant holdout axis.
R7 Inference	Quantified protection → metric, denominator and scoped defended evaluation condition.
R8 Utility	Preserved task/component function → performance at the same protected state/operating point.
R9 Cost	Resource-budget feasibility → units, baseline and cost object at that operating point.

B: bound; **P**: partial binding; U: unreported in inspected scope; A: source unavailable; ?: binding unverified; –: inapplicable. First determine applicability, then assign evidence status. Family **B** means at least one scoped record satisfies the item, not that every result does.

TABLE 12. RECORD-DERIVED EVIDENCE AVAILABILITY FOR ALL 20 EVALUATED-MITIGATION FAMILIES. R1–R9 ARE SCOPED REUSE ITEMS, NOT OVERALL PAPER SCORES.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [5]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	U	–
WRA [6]	**B**	**P**	**B**	–	–	**P**	**B**	**P**	U
NetEcho [40]	**B**	**B**	**B**	**B**	?	**B**	**B**	U	**P**
Early Bird [3]	**B**	**B**	**B**	**B**	?	**B**	**B**	U	**B**
Whisper Leak [43]	**B**	**B**	**B**	**P**	U	**B**	**B**	U	**P**
SpliceLeak [51]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [52]	**B**	**B**	**B**	**P**	U	**B**	**B**	U	**B**
I Know What You Said [23]	**B**	**B**	**B**	–	–	**P**	–	–	**B**
Speculative decoding [42]	**B**	**B**	**P**	**P**	U	**B**	**B**	–	**B**
Dangol workflow [19]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	U	**B**
Cloud traffic benchmark [55]	**B**	**P**	**B**	U	U	**P**	**B**	**P**	U
Remote timing [39]	**B**	**B**	**P**	–	–	**B**	–	U	**B**
TDXRay [30]	**B**	**B**	**B**	–	–	**B**	–	**B**	**B**
SafeKV [46]	**B**	**B**	**B**	–	–	**B**	**B**	U	**P**
AHB [17]	**B**	**B**	**B**	**P**	U	**B**	**B**	**B**	U
KVGov [37]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
CachePrune [7]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [32]	**B**	**B**	**B**	–	–	**B**	**B**	–	**B**
SparLeak [28]	**B**	**B**	**B**	**P**	U	**B**	**B**	**B**	U
Shape and Substance [29]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	20	17	17	4	2	17	16	4	11
Applicable	20	20	20	11	11	20	17	17	19

B: bound; **P**: partial; U: unreported in inspected scope; ?: unverified binding; –: inapplicable. Table 11 defines R1–R9. Each cell is generated from records: B if any is B, otherwise P, ?, U, A, then –. Different columns can refer to different configurations. Table 25 lists every inference/cost projection and its record identities. The 13 prior family-display corrections changed aggregation, not the underlying record judgments (S36).

TABLE 13. QUERY ACCOUNTING FROM THE SAME 33 MITIGATION RECORDS. OUT: AT LEAST ONE REQUIRED ITEM IS INAPPLICABLE; ELIGIBLE: REMAINING RECORDS; BOUND: ALL REQUIRED ITEMS ARE B.

Query	Out	Eligible	Bound	Fam./Cfg.
Inference + utility	11	22	4	3/3
Inference + cost	11	22	5	3/4
Matched adaptation + inference	16	17	2	2/2
Inference + utility + cost	13	20	2	1/1
All five bindings	19	14	0	0/0

Required items by row: R7/R8; R7/R9; R4/R5/R7; R7/R8/R9; all five. The two sets of 22 eligible records differ. Six contextual controls are outside the starting 33. Fam./Cfg. deduplicates bound records. CachePrune supplies the triple join: two outcomes at one configuration. Joins preserve cost scope and do not imply paired effects. Appendix F lists every record membership; Table 14 names all positive rows.

Join evidence at the record, not the family. A family-level B means that at least one record supplies an item. It does not imply that its other B cells concern the same intervention. We therefore code all 351 record–item cells and join only B bindings within the same record and configuration (Table 13). All joint queries begin with 33 mitigation records. A query excludes a record only when a required item is inapplicable; U, ?, and P remain eligible but unbound. Table 13 separates excluded, eligible and bound counts, with deduplicated families/configurations. The six contextual controls are separate. CachePrune supplies the triple join: two outcomes share one quality/cost package. The denominator identifies eligible records for our query, not papers obliged to answer it. SparLeak's intensity-specific ASR/ROUGE pairs supply component utility under R8.

Reading the family and record tables together. Eight family rows have B in both inference and cost, but only three families have those bindings in the same record. Table 14 lists all five positive records; Table 25 shows the eight family projections and their exact supporting records in the same PDF. Early Bird, for example, supplies inference in its prefix-oracle row and cost in its semantic-cache row. CachePrune's two targets instead point to one default package. Appendix F gives all 33 records' membership in all five queries, making the differing denominators visible.

These inventories answer specified feasibility questions. The sole triple package is a useful worked configuration, not a robust estimate of how often the field reports complete evidence. Removing CachePrune removes that example. A published-version-linked subset does not supply another triple package in the inspected records (S36); its small eligible denominators do not support a venue-quality comparison. A new source report can extend the atlas without changing the comparison rules.

SoK Takeaway 2 (Join the operating point). CachePrune's two outcomes identify one reusable inference–utility–cost package. Request evidence for the intended deployment query rather than combine a family's best individual cells. These counts inventory evidence; they do not rank paper quality.

TABLE 14. POSITIVE JOINT RESULTS AND EXACT RECORD IDENTITIES. ALL ROWS START FROM 33 MITIGATION RECORDS; A REQUIRED INAPPLICABLE ITEM EXCLUDES A RECORD. EVERY OUTCOME RETAINS THE SOURCE LOCATOR IN APPENDIX B.

Requested evidence	Bound record IDs	Interpretation at the source configuration
Inference + utility (4/22)	AHB-online-executed; CACHEPRUNE-selective; CACHEPRUNE-context; SPARLEAK-replacement	Executed task outcomes or component quality; the two CachePrune targets share one package.
Inference + cost (5/22)	SPEC-padding-REST; DANGOL-capture; DANGOL-livebinding; both CACHEPRUNE records	Payload overhead, transformed-capture cost, loopback cost and serving cost remain distinct.
Adaptation + inference (2/17)	UI-delay-offline; DANGOL-capture	Matched clean/protected fitting on a defended test; no live-utility inference.
Inference + utility + cost (2/20)	CACHEPRUNE-selective; CACHEPRUNE-context	One family, one default configuration, two target outcomes.
All five items (0/14)	None in this extraction	A specified query has no bound record; no estimate of all published or unpublished evidence.

TABLE 15. THREE EVIDENCE OBJECTS FOR RQ4. THE SAME FAMILY MAY SUPPLY SEVERAL OBJECTS IN SEPARATE EXPERIMENTS.

Object	Required link	Permitted reuse
Attack generalization	Training/profiling source → held-out inputs or systems; attack outcome.	An attacker or recognition method covers the stated target domain.
Defended condition	Named mechanism, local fitting, defended test and outcome.	Retain that operating point; multi-model testing alone is not held-out transfer.
Defense transfer	Source-selected policy/calibration held fixed → target; local-calibration comparator and attacker fit specified.	Transport the evaluated policy only over the tested direction and scope.

6. Which Evidence Supports Defense Transfer?

6.1. Separate the evaluated object before counting holdouts

RQ4 distinguishes three questions: does an *attack* recognize new inputs or systems; is a *defense condition* evaluated; and does a *fixed attacker or defense calibration* carry from a specified source to a target? Table 15 makes the distinction operational. Repeating a defense on several models can establish condition-specific outcomes without establishing a held-out model or transferring a fitted calibration.

The 82-family matrix in Appendix D is a discovery index for these questions. Its six axes are task/input/user, model, harness/topology, hardware, network/load and time. H marks reported separation, V condition variation/recalibration, X a scoped restriction, and ? unresolved extraction. The 13 task/input H entries include only four mitigation families; all three model H entries concern attack/identification generalization in families outside the 20-family mitigation audit. The two topology H entries occur in mitigation families, but their held-out objects are workflow recognizers. These counts therefore cannot be read as 13 task-transfer or three model-transfer validations of defenses. S34 identifies each H entry and its evidence object; S28 preserves the full matrix and source scopes.

6.2. A changed generator and a tested defense are different bindings

LLM-Fin, ArchSnoop and LeakyLMs evaluate recognition of unseen model variants or architectures [79], [27], [50]. This supplies attack-side model generalization. In the traffic benchmark, graph-to-star and star-to-graph classification explicitly move the fitted attacker across topologies; the prompt/temperature countermeasures are separate evaluations [55, V-B–C]. Dangol's leave-composition-out test likewise examines recognition of a novel workflow shape, separately from its protected-capture frontier [19, 9.8–9.12]. Neither topology entry licenses a transported defense calibration.

There is nevertheless useful *defense-side variation* evidence. Time Will Tell applies the same fixed-length instruction to two models and reports different residual classification ASRs; this tests a common rule under changed generators [52, VII-B]. Whisper Leak evaluates mitigation conditions across provider-models, while its prompt holdout describes the classifier protocol [43, 3.2,6]. Preserve those per-condition results; a shared rule, a refitted local attack, a source-trained attack and a source-calibrated defense are four distinct comparisons.

Deployment decision. For the proposed change, name the source, target, held-out unit, quantities held fixed or refitted, and protected outcome. Carry forward only the corresponding binding. An attack-generalization result justifies including that attacker at the target; a defended-condition result supplies its local outcome; a source-to-target defense-calibration result supports only the stated transfer. This prevents both overclaiming portability and discarding useful local evidence. Unresolved hardware/time entries in the discovery matrix identify source follow-up, not absent experiments.

7. Synthesis: From Evidence to Reuse Decisions

The census (§5) covers all 39 records and the holdout matrix (§6) all 82 families. We now select three examples

TABLE 16. CLAIM-CONDITIONED REUSE DECISIONS DERIVED FROM THE RECORD AND HOLDOUT AUDITS. EACH CHANGED BOUNDARY IDENTIFIES THE EVIDENCE THAT MUST BE BOUND BEFORE CARRYING A RESULT FORWARD.

Reuse question	Evidence decision	Retain or measure next
Emitter / representation	An upstream or buffered capture can differ from the final receiving trace and delivery schedule.	Bind inference to the final receiver; retain the emitter's overflow, byte and delivery-delay scope (R2/R3/R9; §7).
Read cover → writes	Discarding a reply does not establish unchanged commits, authorization or workflow outcomes.	Bind the protected state and utility to the executed intervention; include cover calls, quotas and failures (R3/R8/R9).
Customer / model calibration	A reported split can separate inputs without changing the generator or calibration population.	State held-out units and calibration provenance; separate local fitting variability from source/domain changes; evaluate the requested transfer axis (R5/R6; §6).
Attacker observation	An outcome for one projection or fitted attacker does not supply a result for a richer receiving view.	Evaluate comparable defended tests under the intended observation and adaptation contract; retain receiver-specific inference metrics (R2/R5/R7).
Stored trace → online path	A trace transformation or simulated span does not measure executed queues, completion or online latency.	Bind the receiver, harness and resource denominator to the online condition; establish the relevant topology/network holdout (R3/R6/R9).
Bound cache-reuse package	Early Bird prefix recovery and CachePrune direct recovery permit a scoped candidate comparison; CachePrune binds a shared utility/cost package.	Retain that package for its measured configuration; measure prefix-policy serving cost separately, and re-evaluate any changed workload/hardware.
Component → workflow utility	Component quality or logical state conformance does not determine a planner's response to scheduling and deadlines.	Report utility at the requested workflow boundary and the same intervention/configuration as protection and cost (R3/R8/R9).

by the *comparison decision* they expose, not by effect size: a changed receiving path, cover with different state effects, and an adaptation/cost mismatch within a named mechanism. Their value is the decision they change: retain an aligned result, separate a wrongly joined package, or require a result at a changed execution boundary.

7.1. Case 1: bind the final target-bearing emitter

NetEcho's upstream packet-size shaping operates on a different projection from the tool-call sequence. Its plaintext-aware frontend then emits a new size trace (grouped token counts or character lengths), reopening a size channel after upstream protection [40]. SplitDecoy is a separate return-path record: the cloud sees nonzero gradient row support on every tested frame across nine seeds [36, Sections 5.1–5.3]. Both change the receiver boundary and therefore require a separately bound inference result.

The data processing inequality requires $S \to Z \to Y$: a downstream process that receives only Z cannot increase $I(S;Z)$ [80, Theorem 2.8.1]. A frontend also receiving content X instead forms $Y = f(Z, X)$. The applicable bound is $I(S;Y) \leq I(S;Z,X)$; it need not be bounded by $I(S;Z)$. For a constructed binary example, let S be uniform, the protected upstream length Z constant, and the downstream length $Y = 100 + S$ bytes. Then $I(S;Z) = 0$ but $I(S;Y) = 1$ bit and optimal downstream classification is exact. This calculation illustrates the missing conditional-independence premise; it is not a measured leakage estimate for NetEcho.

7.2. Case 2: resolve the WRA versus AHB cover asymmetry

WRA's one-decoy E_{func} drops from 0.770 to 0.610, whereas AHB's Trip Planning success falls from 0.75 to 0.15 under `Online+Fake` humanization [6], [17]. Grouping both as "cover traffic" conceals different objectives and executions.

WRA discards decoy reads for Type I confidentiality; AHB executes GUI events for Type III evasion, potentially navigating or submitting forms. Their utility outcomes are not contradictory, and AHB's missing condition-specific denominators preclude a matched causal effect. A shared workflow setting does not establish shared state effects. Terminal-state reward and dispatch-time authorization also answer different questions: an unconfirmed write can still reach the intended final state. Reuse requires an explicit state/authorization outcome at the intervention being carried forward.

7.3. Case 3: bind adaptive retraining and cost objects

A matched adaptation contrast compares clean-trained D_0 and defense-trained D_2 on identical defended tests. UI Traces and Dangol raise the record-binding question this leaves open: whether a refit, a measured delay and a utility outcome refer to the same executed condition [5], [19]. Cost objects show the same pattern. Early Bird's 6ms/4% GPTCache cost does not price K-prefix isolation; KV-Gov's $1.3\,\mu$s HMAC microcost omits lost reuse; SpliceDefense's QCP cost does not price CTBF+QCP [3], [37], [51]. CachePrune binds protection, QA/summarization utility and throughput, while OTRO separates conditional leakage from

TTFT/padding costs [7], [32]. The reuse decision follows the evaluated configuration: retain each source's bound outcome, and request the missing adaptation, execution or cost measurement before extending it.

7.4. A positive reuse path: from comparison to a candidate configuration

The Early Bird K-prefix record and CachePrune's direct-recovery record form one of the 17 descriptive matches: both address sensitive-token recovery through cache-reuse probes [3, V-A], [7, 5.1–5.6]. This permits an analyst to retain both mechanisms as candidates for that threat, while keeping the oracle and executed-serving measurements separate. The joint audit then changes the next step. CachePrune supplies a default configuration package linking recovery, generation quality and serving cost; the analyst can use that package for a source-scoped feasibility assessment. Early Bird's semantic-cache overhead cannot price its K-prefix candidate, so that candidate still needs a serving-cost measurement. Neither a pooled recovery ranking nor a new-hardware guarantee follows from this comparison.

What the three findings add. Record-level comparison turns a mechanism-level shortlist into correctly scoped alternatives. Configuration-level joint binding distinguishes a reusable package from borrowed utility or cost. Deployment decisions then identify the changed object and the measurement it needs. Compared with established WF practice on adaptive attacks, shifts and online costs [8], [9], the LLM/agent synthesis makes three additional joins explicit: prefix versus semantic-cache primitives, discarded read outputs versus executed state changes, and component generation quality versus planner outcomes. The source-backed examples identify which measurements a particular deployment query may combine and which it must keep separate.

7.5. A deployment query determines the evidence to retain

Start with the intended receiver, protected outcome and intervention. Record-level comparison identifies whether the cited result measures that object. Configuration-level joint binding identifies which inference, utility and cost outcomes share its configuration. Deployment decisions use the holdout matrix to identify evidence for the proposed change and specify the remaining measurement. Table 16 makes these outputs operational: a new emitter needs a receiver-specific capture, writes need state/authorization checks, source-calibration comparisons need a local fitting-variability control, and an online scheduler needs its own planner utility and latency measurement. The calibration control follows from the comparison design: changing the source can also change the finite training sample used to fit the defense.

The following worked applications show how the procedure represents external reevaluations and internal examples. Appendix E retains the detailed controlled case study.

TABLE 17. EXTERNAL APPLICATIONS: RETAIN A SCOPED RESULT, SEPARATE AN OUTCOME PACKAGE, OR REQUEST A SPECIFIC MEASUREMENT. FULL 15-RECORD LEDGER: S35. ACCURACY AND TPR RETAIN THEIR ORIGINAL POPULATIONS.

Published contrast	Evidence-reuse decision
WTF-PAD [81]	Refit: DF accuracy 90.7%; prior attackers in the same table reach at most 69.0%.
Walkie-Talkie [81]	Retain symmetric-pairing scope: DF accuracy 49.7%; do not inherit it for asymmetric pairing.
FRONT [82]	Separate outcomes: live DF TPR improves, 72.06% to 42.79%, while data overhead rises, 52% to 75%.
Tamaraw [82]	Retain low live TPR, 6.43%; use live bandwidth overhead 179%, rather than simulated 207/224%.
DynaFlow [9]	Separate live record: 379% mean per-site time overhead; changed URLs/crawler prevent a matched simulation–live effect.

7.6. External worked applications: retain, separate, remeasure

We apply the procedure to a fixed external frame: both defenses in Deep Fingerprinting, all three in WFDefProxy, and all nine in Mathews et al.'s reevaluation, with DynaFlow's live implementation kept as a separate record [81], [82], [9]. The resulting 15 measurement records represent 14 defense–study pairs in three studies. The ledger retains favorable outcomes alongside weaknesses, fixes the Standard setting for the nine-defense table, and records a locator and a concrete measurement decision for every row (S35).

The worked output separates decisions that a mechanism name conflates (Table 17). Stronger protected-data fitting requires revisiting WTF-PAD's attack result, while Walkie-Talkie's symmetric-pairing result remains a valid scoped outcome. FRONT's live protection improves even as its bandwidth cost changes. A record-level audit therefore preserves the security observation and replaces the cost estimate separately. DynaFlow's live result needs a separate execution/workload record, preventing a simulated cost from being attached to a live attack result.

The complete frame produces 11 protected-fit/view measurement requirements and four execution/cost requirements. Four favorable controls remain usable in their measured scopes: symmetric Walkie-Talkie, live FRONT, live Tamaraw and one-split TrafficSliver. The actionable output is a scoped retain, separate or remeasure decision for each record. The examples faithfully encode established WF practice. Outcomes were visible during extraction, so agreement is an illustration of the representation rather than evidence of prediction accuracy or an improvement over ordinary expert reading. These three WF studies do not establish LLM/agent effectiveness; the cache and tool examples provide the domain-specific application.

7.7. Seven internal illustrative judgments

The existing controlled study supplies illustrative applications: seven written judgments specify which apparent

TABLE 18. CURRENT OUTCOMES OF ALL SEVEN TIME-ORDERED INTERNAL JUDGMENTS. SUPPORTED REFERS TO THE SPECIFIED COUNTEREXAMPLE, NOT A FRAMEWORK-ACCURACY ESTIMATE.

ID	Reuse claim tested	Finding and next measurement
P1	Buffered protection/cost carries to a stream	Supported: verbosity accuracy 100%; capture the stream emitter.
P2	Low clean-trained accuracy establishes protection	Supported: D_0 50%, D_2 100%; refit on protected data.
P3	Padded events hide response identity	Supported: top-1 58–62%, chance 2.1%; test the whole trace.
P4	Equal target-verb count hides cancellation	Supported: count BA 50%, suite BA 76.9%; expand the view.
P5	Protected dispatch implies a protected tool link	Source premise unmet; excluded from favorable evidence. Establish dispatch protection first.
P6	Framing alone hides intent	Partial: bucketed return leaks; resolve the remaining target cells.
P7	Joint cover/framing protects both targets	Not established: retain the joint screen before accepting protection.

protection claims should survive a changed target, receiver or attacker. The protocol-only repository state predates the first archived result commit on 21 October 2026; S35 preserves the ordering receipt, original labels and current reassessment. Table 18 brings all seven into the main paper.

Four judgments have supporting counterexamples; one is partly supported. P5 has a failed source premise and is excluded from favorable evidence for either transfer failure or framework success. P7's proposed joint protection remains unestablished. The examples specify the receiving view and protected-data fit used in each calculation. Repository ordering establishes chronology only; author-designed examples do not measure the accuracy of the procedure. The decisions differ by failure mode: P1–P4 require the changed emitter, attacker or trace view to be measured; P5 requires establishing source protection first; P6/P7 require evidence for the still-open claim. Appendix E retains the measurement scopes and statistical qualifications.

8. Implications and Ethics

8.1. Deployment and reporting implications

The decision table distinguishes receiving permissions from intervention capabilities. An access-network operator, VPC monitor and receiving tool service expose different views; remote paths can reveal destinations even with Encrypted ClientHello [83]. A result for one ingress projection therefore needs a separately bound claim before it is applied to host instrumentation or downstream commit logs.

Likewise, read queries, provider-isolated execution and gateway-routed cover have different state contracts. Kubernetes dry-run, Stripe sandboxes and transactional rollback offer isolation mechanisms with their own limits [84], [85], [86]; idempotence alone is insufficient, and sequence changes can survive rollback [87]. Evidence reuse requires the trusted boundary, authorization policy, resource quotas and protected outcome to accompany the mechanism name. The corpus's component outcomes remain useful when their execution and cost scopes match the intended query.

TABLE 19. REUSABLE REPORTING CARD. FILL EACH EVIDENCE CELL WITH A VERSION AND LOCATOR OR "UNKNOWN"; THE FINAL ACTION IS CONDITIONAL ON THE PROPOSED DEPLOYMENT.

Field	What to enter
Receiver/target	Hidden property, accessible signal and permissions.
Intervention	Primitive, execution mode, state and trust boundary.
Attacker/test	Training or calibration condition; defended test population.
Function	Component quality or task outcome; shared configuration.
Cost	Resource, units, denominator, baseline and included operations.
Extension	Changed workload/system axis; available holdout; retain, separate or measure next.

A usable reporting card. Table 19 is a reusable six-field template. Its last column asks for a deployment decision after recording the source evidence. Table 20 fills it for two venue-linked sources and the running cache example, preserving their different scopes. The editable artifact template includes an explicit unknown state and source-locator slot for every field.

8.2. Ethics and artifact scope

The audit uses source-linked public research records. Stakeholders include tenants, shared-capacity providers and users affected by observable behavior or state changes. Bot-detection evasion and research-intent privacy have different authorization relations [17], [6]. The appendix case study uses cloned public benchmark databases and controlled sockets; model interactions use public tasks and simulated users. No production account is modified.

AI tools assisted candidate-field extraction, implementation, source reconciliation and prose revision. The author-review coverage is stated in §2. Reliability controls retain source versions and locators, separate unavailable from unreported evidence, preserve corrections, and recompute aggregation and joins from the record cells before compilation. These controls detect inconsistent accounting and support source inspection; they do not estimate an AI extraction-error rate. Authors retain responsibility for source interpretation. Case-study observations and protocols remain separate from the literature atlas.

9. Conclusion

This SoK organizes defense evidence around the changes a deployment introduces: a different observed signal, executed intervention or evaluation setting. The record atlas supports scoped comparisons, identifies outcomes measured at the same configuration, and turns holdout evidence into concrete measurement plans. Its practical output is visible in the cache example: retain prefix-recovery candidates for

TABLE 20. FILLED REPORTING CARDS FROM SOURCE RESULTS. EARLY BIRD IS VENUE-LINKED (INSPECTED AUTHOR VERSION); TDXRAY IS AN S&P PAPER; CACHEPRUNE IS A CITED PREPRINT AND SERVES AS THE RUNNING CONFIGURATION EXAMPLE. NO NEW BENCHMARK IS IMPLIED.

Field	Early Bird [3, V-A–B]	TDXRay [30, 6.1, Fig. 8, Table 3]	CachePrune [7, 5.1–5.6]
Receiver/target	Timing oracle; sensitive prefix recovery.	Memory-access view; oblivious lookup defense component.	Cache-reuse probes; direct and contextual token recovery.
Intervention	Prefix-granularity oracle; semantic-cache prototype separately.	Oblivious mapping microbenchmark.	Selective cache sharing at default configuration.
Attacker/test	Granularity-specific calibration; matched adaptation unresolved.	Component cost, no bound protected end-to-end inference row.	Recovery evaluation; default quality/cost package; no matched adaptation contrast.
Function	No bound prefix-policy utility.	No bound full-inference utility.	QA/summarization quality at stated configuration.
Cost	Semantic-cache overhead belongs to a different primitive.	Lookup microcost; not complete serving cost.	Serving throughput/cost; preserve default package.
Extension/action	Retain prefix candidate; measure its own serving cost.	Retain component cost; measure protected service and its function.	Retain scoped package; test changed workload/hardware.

a shared threat, keep semantic-cache cost separate, and carry forward a quality/cost package only at its measured operating point.

The published record identities, query memberships and filled reporting cards make these decisions inspectable. Pair frequencies depend on the indexing rules; the scientific value lies in the source-supported distinctions and reusable decision process. Worked applications illustrate that process while preserving the scope and uncertainty of each source result.

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Appendix A. Audit Trail

The artifact preserves the 82-family map, 39-record ledger, 180 family cells and 351 record cells with source versions and locators. S33 publishes query membership and sensitivity checks. Case-study protocols, observations and uncertainty calculations retain their original cohorts and provenance.

Appendix B. Result-record index

Table 21 indexes all 39 records by measurement mode and source locator. Marks distinguish six contextual controls (‡) and six final-search additions (†); all enter the pair inventory. The ledger retains full locators, versions and hashes.

Appendix C. Case-Study Artifact Index

Table 22 maps the single appendix case to historical artifact keys and supplementary methods. These keys preserve original task/customer memberships; S30 retains the corrected permutation/Holm tests and calibration resampling.

Appendix D. Family-Level Holdout Evidence

This is a family-level discovery index, not a count of defense-transfer tests. The three model H entries concern attack/identification generalization; even an H in a mitigation family must be linked to its defended condition before reuse. The main RQ4 table separates those evidence objects, and S34 identifies every H entry. Table 23 covers 82 families and 492 cells. Axes: T task/input/user; M model; C harness/topology; H hardware; N network/load; D time. Codes: H reported holdout; V condition comparison/recalibration; X scoped restriction; ? unresolved; – conceptual/inapplicable. Scope C/A denotes core/adjacent. H records separation, not transfer success; ? does not establish an absent experiment. S28 retains source locators and qualified interpretations.

Appendix E. Illustrative Case Study and Reconstruction Scope

This appendix explains the role of the archived implementation examples. They instantiate the same distinctions used in the literature atlas: final receiver versus upstream trace, executed state versus discarded responses, and online planner utility versus reused trajectories. They are illustrative applications, not validation data for the comparison rules. The complete numerical grids, negative outcomes, intervals and corrected tests remain unchanged in S11–S32.

The transport example has 48 saved responses on 24 topics (12 train/12 test). The public *retail* setting has 115 tasks from 53 customers; its frozen split contains 46 test tasks from 24 customers, including 5 cancellation-positive and 7 return-positive customers. The separate live planner example has 100 episodes: 37/50 baseline and 33/50 defended successes, a difference of -8.0 points with customer interval $[-23.9, 6.5]$. All eight calibration contrasts have Holm-adjusted $p = 1.000$; the original absolute protection screen remains unmet. Neither these unresolved contrasts nor resampling the same customers validates the framework.

The artifact separates *analysis reconstruction* from *new model collection*. Archived requests, responses, traces, calibration hashes and predictions permit the reported calculations to be reconstructed. Fresh calls to mutable Sol/Luna/MiniMax/GLM aliases, with documented unequal budgets, cannot reproduce an identical generator. No fixed-weight replication claim is made. Historical protocols and model identities remain explicit in S14–S32; the literature synthesis does not depend on obtaining a favorable result from these examples.

Appendix F. Reconstructing the Audit in This PDF

Table 24 provides every mitigation record's query membership, including inapplicability. Table 25 explains every family-level inference/cost conjunction without importing evidence between records. The 39 record IDs, measurement modes and source locators are in Appendix B; Table 9 covers all 133 declared pair decisions. The released pair CSV lists individual source locators and reasons. Thus the core denominators and positive joins can be checked in this PDF; the supplement supplies expanded rationale and historical corrections.

TABLE 21. COMPLETE INDEX OF THE 39 RESULT RECORDS (26 FAMILIES). MODE IS THE PRIMARY MEASUREMENT MODE: EX EXECUTED COLLECTION/PROTOTYPE (20), TR TRACE TRANSFORMATION (6), SI SIMULATION/DERIVED ESTIMATE (5), CO COMPONENT/CALIBRATION MEASUREMENT (6), UN PLACEMENT UNRESOLVED (2). ‡: CONTEXTUAL RECORD (6); †: ADDED BY THE FINAL SEARCH (6). T=TABLE, F=FIGURE, APP.=APPENDIX; LOCATORS REFER TO THE CITED SOURCE.

Record ID / family	Mode	Boundary; source locator
UI-delay-offline [5]	Tr	Offline delay; no live task outcome. 6.1 F5; A.5
WRA-decoy-execution [6]	Ex	Executed decoys; usefulness proxy only. 8.1–8.3 T8–11
NET-upstream-local [40]	Ex	Protected deployment/local serving; topology scoped. IV-C–E TIII–IV
NET-dummy-padding-simulation [40]	Tr	Synthetic insert/pad; no live execution. VII TIX
EARLY-K-calibration-oracle [3]	Si	K-specific fit then oracle; not live KV trial. V-A TVI; IV-A
EARLY-semantic-deidentification [3]	Ex	GPTCache prototype cost; not KV-granularity cost. V-B
WHISPER-randompadding-capture [43]	Ex	New padded captures; rounded medians. 6.1 T3–4
WHISPER-batching-replay [43]	Tr	Offline batching; model-dependent result. 6.2 F7
WHISPER-injection-replay [43]	Tr	Synthetic packets; no live zero-delay claim. 6.3 T5
SPLICE-CTBF-signal [51]	Ex	Prototype background load; tested signal only. VII-C TII
SPLICE-QCP-cost [51]	Co	QCP component cost; excludes combined CTBF. VII-C TII
TIME-length-instruction [52]	Ex	Length instruction; unequal outputs remain. 7.2
TIME-padding-estimate [52]	Si	Analytic plaintext cost; not network bytes. 7.3
SPEC-padding-REST [42]	Un	Padding/aggregation; payload cost, placement unverified. 6.1 F6–7 T2
DANGOL-capture [19]	Tr	Capture transform; distinct from live population. 9.8–9.9 T6
DANGOL-livebinding [19]	Ex	Live loopback binding; do not join capture accuracy. 9.7, OHTTP paragraph after T5
SIG-common-guardrail‡ [59]	Ex	Common instruction; not a causal guardrail effect. 4.1–4.2; App. C
SAFE-post-trigger [46]	Ex	Post-trigger probes; detector cost not service TTFT. 7.2.4 F12; 7.3–7.4
AHB-online-executed [17]	Ex	Executed UI edits; task/state changes possible. 5.1.2 T1; E.1–E.3
AHB-offline [17]	Tr	Offline touch edits; no online task damage claim. T1 offline rows; E.1–E.3
KVGOV-partialflush-sim [37]	Si	Deterministic simulation; not live protected recovery. 5.1–5.3
KVGOV-hardware-gap [37]	Co	Hardware gap calibration; not protected ASR. 5.7–5.8
KVGOV-HMAC-cost [37]	Co	HMAC microcost; not cache reuse/task loss. 5.9
TRAFFIC-prompt-reformulation [55]	Ex	Collected prompt rewrites; task utility unmeasured. V-C1 F9a
TRAFFIC-temperature [55]	Ex	Collected decoding changes; no monotone benefit. V-C1 F9a
CARLINI-fixed-rate [39]	Un	Rate/replayed cost; per-token, not request delay. VII F11–12
GAO-zero-copy-off [23]	Co	Loader component costs; not end-to-end TTFT. 7.1
TDXRAY-oblivious-map [30]	Co	Lookup microbenchmark; not full inference cost. 6.1 F8 T3
AGENTTELL-fixed-instruction‡ [66]	Ex	Live local browser tasks; loaded/cold context contrast. 4.3.1; 5 Finding 1
FP-CLOUDFLARE‡ [67]	Ex	Live access control; not user confidentiality. 6.4 T5
BAR-co-running-noise‡ [24]	Ex	MIG noise test; not an LLM defense trial. 7.2–7.3 T8
RHYTHM-VPN‡ [38]	Ex	Live VPN shift; baseline-trained attacker only. VI-B; VII
DEB-filter-origin‡ [47]	Ex	Filter creates attack view; not mitigation of it. Threat model; 4.2
CACHEPRUNE-selective† [7]	Ex	Selective cache sharing; direct token recovery/quality. 3.1; 5.1–5.6; T1; App. A
CACHEPRUNE-context† [7]	Ex	Contextual guessing; distinct from direct recovery. 3.1; 5.1–5.6; T1; App. A
OTRO-unpadded† [32]	Co	ORAM TTFT/memory; access-count padding excluded. VII-C–E; TIII
OTRO-epochpad† [32]	Si	Epoch-padding estimate; not measured padded TTFT. VII-C–E; TIII
SPARLEAK-replacement† [28]	Ex	Sparse replacement/quality; attack cost is separate. 3.1; 4–7; T4
SHAPE-padding-estimate† [29]	Si	Constant-work proposal/estimate; no executed pair. IX-B1

TABLE 22. ARTIFACT INDEX FOR THE APPENDIX CASE STUDY. COMPONENT NAMES DESCRIBE THE ARCHIVED MEASUREMENT BOUNDARY; THE RIGHT COLUMN LOCATES IMPLEMENTATION AND FULL-GRID DETAIL.

Case-study component	Artifact keys	Observation/execution boundary and comparison	Detailed source
Transport emission	E1, E3, E5	Buffered two-link re-emission and token-stream emitters; upstream framing, final-link framing and matched attacker refitting are distinct comparisons.	S9/S11; S12
State and link	E2, E6, E7	Prescribed stateful plans and randomized dispatch cover, then agent–tool TLS composition; dispatch-visible calls and encrypted-link records are separate views.	S9/S11; S12
Trajectories and localization	E9, E10, E12	Saved autonomous Sol trajectories; original pre-test calibration, exploratory customer cross-fitting and oracle operation-category localization retain their different fit/test boundaries.	S14; S15/S21
Timing and WF controls	E13, E14	Local live tool-loop timing/full tunnels, then stronger learners and specified WF adaptations; measured clocks and simulations are distinguished, with matched suites for joint comparisons.	S19; S20
Extensions	E8	Stand-in-planner pipeline check.	S13
	E11	Retrospective training-fold operating-point selection with the historical suite.	S15
	E15	Additional same-provider Luna cohort.	S22
	E16	Same-operation semantic-attribute diagnostic on reused Sol data.	S23.6
	R11/R12	Two served model endpoints: native trajectories, frozen training calibration and TLS write localization.	S24
	R14	Attacker/calibration transfer matrices, non-oracle write detection and write-only framing with the endpoint suite.	S25
	R15	Fresh paired live-planner episodes under isolated prefix cover and injected tool-response deadlines.	S26

TABLE 23. COMPLETE HOLDOUT-CODE MATRIX; CODES AND AXES ARE DEFINED IN APPENDIX D. SOURCE-QUALIFIED ROW NOTES ARE IN SUPPLEMENT S28.

Family	Scope	T	M	C	H	N	D	Family	Scope	T	M	C	H	N	D
F01 AgentPrint [4]	C	?	?	?	?	?	?	F42 ML privacy channels [47]	C	?	V	?	?	?	?
F02 UI Traces [5]	C	V	V	V	?	?	?	F43 MoE stealing [53]	C	?	?	?	?	?	?
F03 WRA [6]	C	H	?	?	?	?	?	F44 Energon [21]	C	?	V	?	?	?	?
F04 FP-Agent [67]	C	H	?	X	?	?	?	F45 SafeKV [46]	C	?	V	?	?	?	?
F05 Prompt length [1]	C	H	?	V	?	V	V	F46 InferNet [88]	A	?	V	?	V	?	?
F06 PromptPeek [2]	C	V	?	?	?	V	?	F47 ArchSnoop [27]	C	?	H	?	V	?	?
F07 NetEcho [40]	C	V	?	V	?	?	?	F48 Coding fingerprints [12]	A	X	?	X	?	?	X
F08 Early Bird [3]	C	?	V	?	?	?	?	F49 AHB [17]	C	?	?	X	?	?	?
F09 InputSnatch [49]	C	H	?	?	?	?	?	F50 WildAuth [13]	A	H	?	X	?	?	?
F10 Whisper Leak [43]	C	H	V	?	?	?	?	F51 Cycles [58]	C	X	?	X	?	?	X
F11 Spill the Beans [26]	C	V	?	?	?	?	?	F52 LumiMAS [57]	C	?	V	V	?	?	?
F12 SpliceLeak [51]	C	V	?	?	?	?	?	F53 SymDiag [89]	A	V	V	?	?	?	?
F13 Time Will Tell [52]	C	?	?	?	?	V	V	F54 LeakyLMs [50]	C	?	H	?	V	?	?
F14 Prompt Echo [41]	C	H	V	?	?	?	?	F55 KVGov [37]	C	?	?	?	V	?	?
F15 I Know What You Said [23]	C	H	V	V	V	?	?	F56 Data-centric survey [90]	A	-	-	-	-	-	-
F16 Speculation [42]	C	H	?	?	?	V	?	F57 Serving RCA [35]	C	?	?	X	?	?	?
F17 Behind Bars [24]	C	?	V	?	V	?	?	F58 ZeroPrint [14]	A	?	V	?	?	?	?
F18 Prompt caching [48]	C	?	V	?	?	?	V	F59 Endpoint fingerprints [15]	A	?	?	V	?	?	V
F19 Whose Agent [65]	C	?	?	?	?	?	?	F60 StepFinder [16]	A	H	?	?	?	?	?
F20 Workflow traffic [19]	C	?	?	H	?	?	?	F61 Web-agent detection [56]	C	?	V	?	?	?	?
F21 GPU contention [44]	C	?	?	V	?	V	?	F62 Cross-campaign [63]	C	V	?	V	?	?	?
F22 AgentTell [66]	C	V	?	?	?	V	?	F63 AgentProv [61]	C	?	V	?	?	?	?
F23 Traffic benchmark [55]	C	?	?	H	?	?	?	F64 LiDAR [62]	C	?	V	V	?	?	?
F24 Remote timing [39]	C	?	V	?	?	?	V	F65 LLM-WFIN [34]	C	V	?	?	?	?	?
F25 Persona fingerprint [64]	C	?	?	X	?	?	?	F66 MIRAGE [54]	C	V	V	?	?	?	?
F26 TDXRay [30]	C	?	?	?	?	?	?	F67 Routing metadata [45]	C	V	?	V	?	?	?
F27 Detokenization [33]	C	?	?	X	?	?	?	F68 Shape and Substance [29]	C	V	V	V	V	?	?
F28 Spillage [91]	A	V	?	?	?	?	?	F69 SplitDecoy [36]	C	?	?	?	?	?	?
F29 PlanTwin [92]	A	V	?	?	?	?	?	F70 PAGE404 [25]	C	V	V	?	?	?	?
F30 SICC [93]	A	?	?	?	?	?	?	F71 Edge EM [20]	C	H	?	?	?	?	?
F31 TOPR [94]	A	V	?	?	?	?	?	F72 SparLeak [28]	C	V	V	V	?	?	?
F32 ClawGuard [18]	C	?	?	V	?	?	V	F73 Sparseety [31]	C	V	V	V	?	?	?
F33 SIGLEAK [59]	C	V	?	?	?	?	?	F74 OTRO [32]	C	?	?	?	?	?	?
F34 ProcGrep [60]	C	H	?	V	?	?	?	F75 CachePrune [7]	C	V	V	?	?	?	?
F35 SecEmb [95]	A	?	X	?	?	?	?	F76 Privacy foundations [70]	A	-	-	-	-	-	-
F36 Mobile agents [96]	A	?	?	?	?	?	?	F77 DP-MemView [97]	A	?	?	?	?	?	?
F37 Hidden risks [98]	A	?	?	?	?	?	?	F78 PriCon [99]	A	?	?	?	?	?	?
F38 Browser privacy [100]	A	?	?	V	?	?	?	F79 SCAAR [101]	A	?	?	?	?	?	?
F39 LLMs Have Rhythm [38]	C	?	V	?	?	H	V	F80 CacheSolidarity [102]	A	?	?	?	?	?	?
F40 MoEcho [22]	C	H	V	?	?	V	?	F81 Vision encoder [103]	A	?	?	?	?	?	?
F41 LLM-Fin [79]	A	?	H	?	?	?	?	F82 Split gradients [104]	A	?	?	?	?	?	?

TABLE 24. ALL 33 MITIGATION RECORDS: QUERY MEMBERSHIP. B: ELIGIBLE AND BOUND; E: ELIGIBLE WITH AT LEAST ONE INCOMPLETE BINDING; X: REQUIRED ITEM INAPPLICABLE. IU=INFERENCE/UTILITY; IC=INFERENCE/COST; AI=ADAPTATION/INFERENCE; IUC=TRIPLE; ALL=R4/R5/R7/R8/R9. FULL R1–R9 SOURCE RATIONALES REMAIN IN S28.

Record / cited family	IU	IC	AI	IUC	All
UI-delay-offline [5]	E	x	B	x	x
WRA-decoy-execution [6]	E	E	x	E	x
NET-upstream-local [40]	E	E	E	E	E
NET-dummy-padding-simulation [40]	E	E	E	E	E
EARLY-K-calibration-oracle [3]	E	E	E	E	E
EARLY-semantic-deidentification [3]	x	x	x	x	x
WHISPER-randompadding-capture [43]	E	E	E	E	E
WHISPER-batching-replay [43]	E	E	E	E	E
WHISPER-injection-replay [43]	E	E	E	E	E
SPLICE-CTBF-signal [51]	E	E	x	E	x
SPLICE-QCP-cost [51]	x	x	x	x	x
TIME-length-instruction [52]	E	E	E	E	E
TIME-padding-estimate [52]	x	x	x	x	x
SPEC-padding-REST [42]	x	B	E	x	x
DANGOL-capture [19]	E	B	B	E	E
DANGOL-livebinding [19]	E	B	E	E	E
SAFE-post-trigger [46]	E	E	x	E	x
AHB-online-executed [17]	B	E	E	E	E
AHB-offline [17]	E	x	E	x	x
KVGOV-partialflush-sim [37]	E	E	x	E	x
KVGOV-hardware-gap [37]	x	x	x	x	x
KVGOV-HMAC-cost [37]	x	x	x	x	x
TRAFFIC-prompt-reformulation [55]	E	E	E	E	E
TRAFFIC-temperature [55]	E	E	E	E	E
CARLINI-fixed-rate [39]	x	x	x	x	x
GAO-zero-copy-off [23]	x	x	x	x	x
TDXRAY-oblivious-map [30]	x	x	x	x	x
CACHEPRUNE-selective [7]	B	B	x	B	x
CACHEPRUNE-context [7]	B	B	x	B	x
OTRO-unpadded [32]	x	x	x	x	x
OTRO-epochpad [32]	x	E	x	x	x
SPARLEAK-replacement [28]	B	E	E	E	E
SHAPE-padding-estimate [29]	E	E	E	E	E

TABLE 25. EVERY FAMILY WITH B IN BOTH INFERENCE AND COST COLUMNS OF TABLE 12. SEPARATE COLUMN WITNESSES DO NOT AUTOMATICALLY FORM ONE ROW-LEVEL RESULT. IDS EXPAND IN APPENDIX B.

Family	Inference record(s)	Cost record(s)	Same-record inference + cost
[3]	EARLY-K-calibration-oracle	EARLY-semantic-deidentification	–
[51]	SPLICE-CTBF-signal	SPLICE-QCP-cost	–
[52]	TIME-length-instruction	TIME-padding-estimate	–
[42]	SPEC-padding-REST	SPEC-padding-REST	SPEC-padding-REST
[19]	DANGOL-capture; DANGOL-livebinding	DANGOL-capture; DANGOL-livebinding	DANGOL-capture; DANGOL-livebinding
[37]	KVGOV-partialflush-sim	KVGOV-HMAC-cost	–
[7]	CACHEPRUNE-selective; CACHEPRUNE-context	CACHEPRUNE-selective; CACHEPRUNE-context	CACHEPRUNE-selective; CACHEPRUNE-context
[32]	OTRO-epochpad	OTRO-unpadded	–